

Lab 3: Data Wrangling and Probability

STAT 141, Week 3

Your name here

In this lab, you will - Practice using data wrangling functions - Practice applying probability rules to answer questions about data

Add any collaborators here!

I collaborated with...

Remember that you are not allowed to use any internet resources or generative AI models (like ChatGPT or Claude) on your assignment. While there are many ways to achieve tasks in R, you may only use packages and functions that have been covered in this course. If you are unsure of how to approach an exercise or of what function to use, help is available to you! Please visit office hours, collaborate with your peers, or post in the Slack channel.

```
# Load the necessary packages
library(lubridate)
library(tidyverse) # Includes ggplot2 and dplyr
```

Exercises

Exercise 1: COVID-19 Case Fatality Rates

We will use data from a 2021 paper published by IEEE, [von Kugelgen J et al. 2021](#). This paper analyses case fatality rates of COVID-19 in Italy and China during 2020.

a. Run the chunk below to view data on COVID-19 cases and fatalities by age group in Italy (on March 9th) and China (on February 22). What does a row represent? [Text only for this question]

```
covid <- read.csv("data/italy_china_covid_comp.csv")
print(covid)
```

	country	age	cases	fatalities
1	China	10-19	549	1
2	Italy	10-19	85	0
3	China	20-29	3619	7
4	Italy	20-29	296	0
5	China	30-39	7600	18
6	Italy	30-39	470	0
7	China	40-49	8571	38
8	Italy	40-49	891	1
9	China	50-59	10008	130
10	Italy	50-59	1453	3
11	China	60-69	8583	309
12	Italy	60-69	1471	37
13	China	70-79	3918	312
14	Italy	70-79	1785	114
15	China	80+	1408	208
16	Italy	80+	1532	202

b. Write R code to *filter* to only the rows corresponding to China using the `filter()` function. Be careful not to overwrite the original `covid` data frame. [Code only for this question]

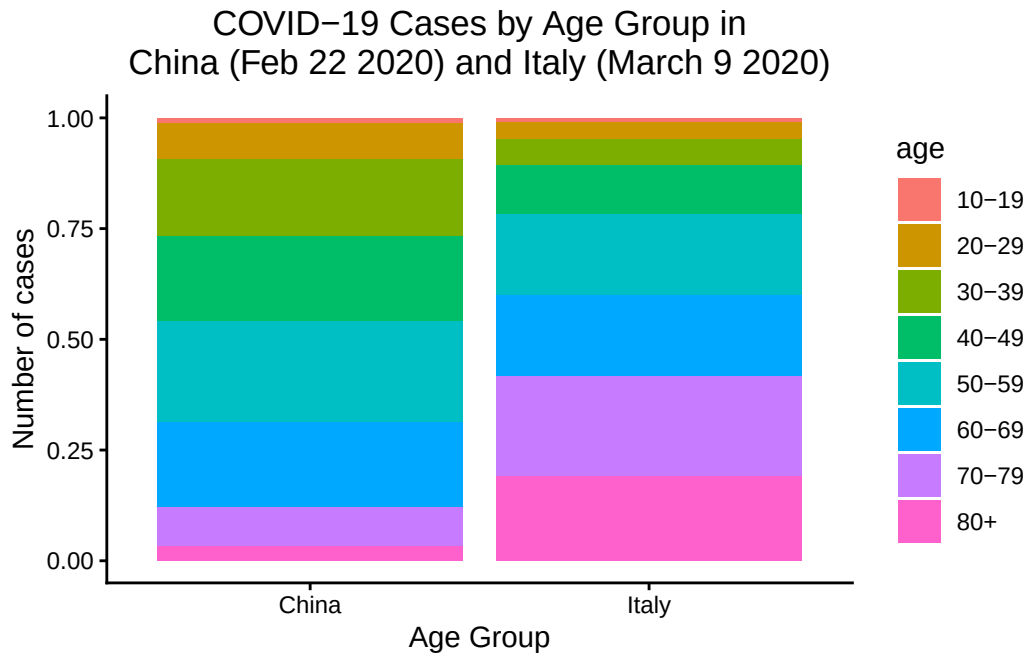
c. Using R in a new chunk below, *create a new column* in `covid` using `mutate()` that calculates the case fatality rate (the proportion of cases that resulted in fatality) for each of the 16 rows. What do you notice when comparing rates between the two countries? [Text and code for this question]

d. Using R in a new chunk below, now calculate the overall case fatality rate within each country (not broken down by age). *This will require you to chain together two **dplyr** functions: `group_by()` and `summarize()`.* How do the overall case fatality rates compare between the two countries? Is this what you would expect based on part (c)? [Text and code for this question]

e. These results are an example of **Simpson's Paradox**, a phenomenon where trends observed in aggregate comparisons reverse when we control for other variables. Run the chunk below to look at a barplot showing the number of COVID-19 cases per age group. How does this help explain what you observed in parts (c) and (d)?

```
covid <- read.csv("data/italy_china_covid_comp.csv")

ggplot(covid, aes(x = country, y = cases, fill = age, label = cases)) +
  geom_col(position = "fill") +
  xlab("Age Group") +
  ylab("Number of cases") +
  ggtitle("COVID-19 Cases by Age Group in\nChina (Feb 22 2020) and Italy (March 9 2020)") +
  theme_classic() +
  theme(plot.title = element_text(hjust = 0.5))
```



Exercise 2: Some Interesting Survey Results

In a previous year, students in Stat 141 were asked two questions as part of a larger survey: “Are hot dogs sandwiches?” and “If dogs wore pants, would they wear them on their front legs, back legs, or on all four legs”? A summary of responses for 70 students is given below:

	Hotdog is a sandwich	Hotdog is not a sandwich
Front legs	1	1
Back legs	17	26
All 4 legs	9	16

Suppose we randomly choose 1 student who completed this survey.

- a.** Are the events “the student thinks a hotdog is not a sandwich” and “the student thinks dogs should wear pants on their back legs” *mutually exclusive*?

b. What is the probability that the randomly chosen student thinks a hotdog is a sandwich?

c. What is the probability that the randomly chosen student thinks a hotdog is a sandwich *and* thinks that dogs should wear pants on all 4 legs?

d. What is the probability that the randomly chosen student thinks a hotdog is a sandwich *given* that the student thinks dogs should wear pants on all 4 legs?

e. Is the event “the student thinks a hotdog is a sandwich” *independent* (in the statistical sense) of the event “the student thinks dogs should wear pants on their front legs”? Explain your reasoning. Do your statistical findings match your intuition about whether these events are related?
